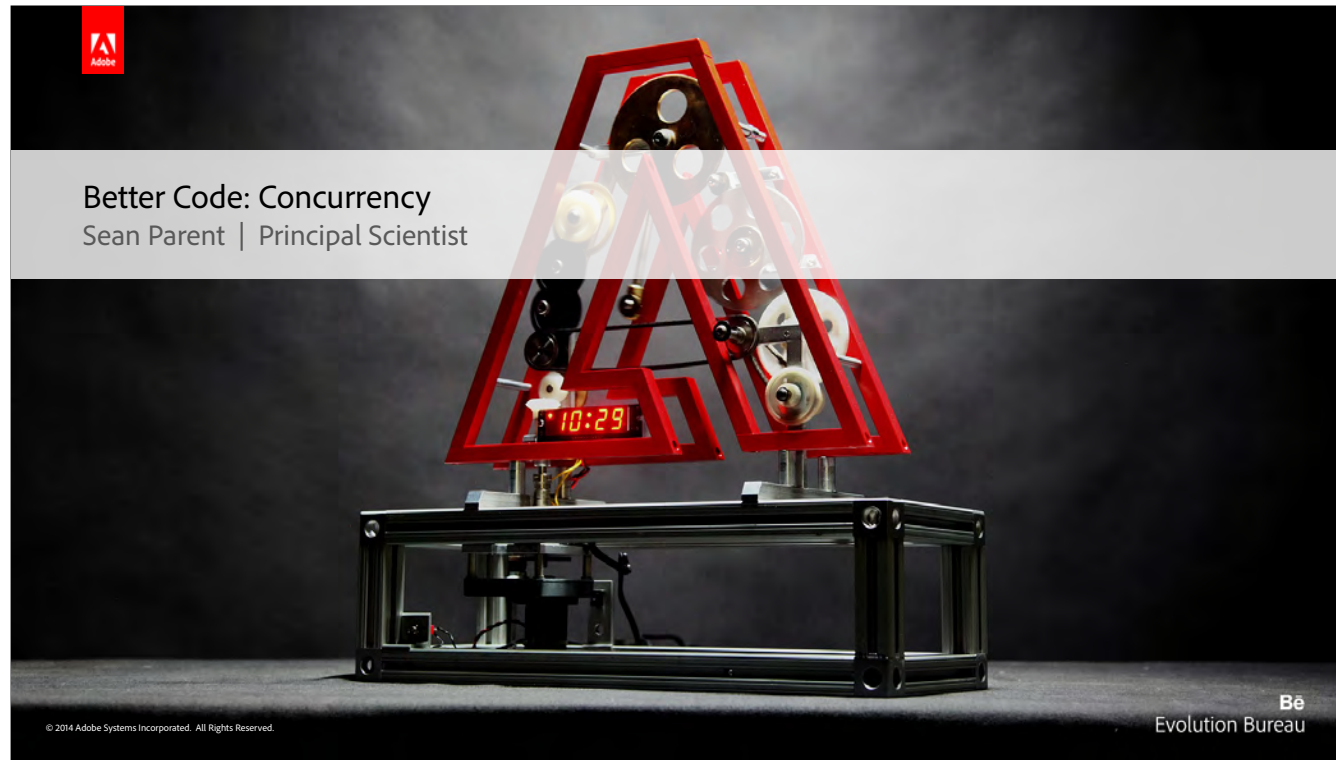




Self Intro

Better Code

- Regular Type
 - Goal: Implement Complete and Efficient Types
- Algorithms
 - Goal: No Raw Loops
- Data Structures
 - Goal: No Incidental Data Structures
- Runtime Polymorphism
 - Goal: No Inheritance
- Concurrency
 - Goal: No Raw Synchronization Primitives
- ...

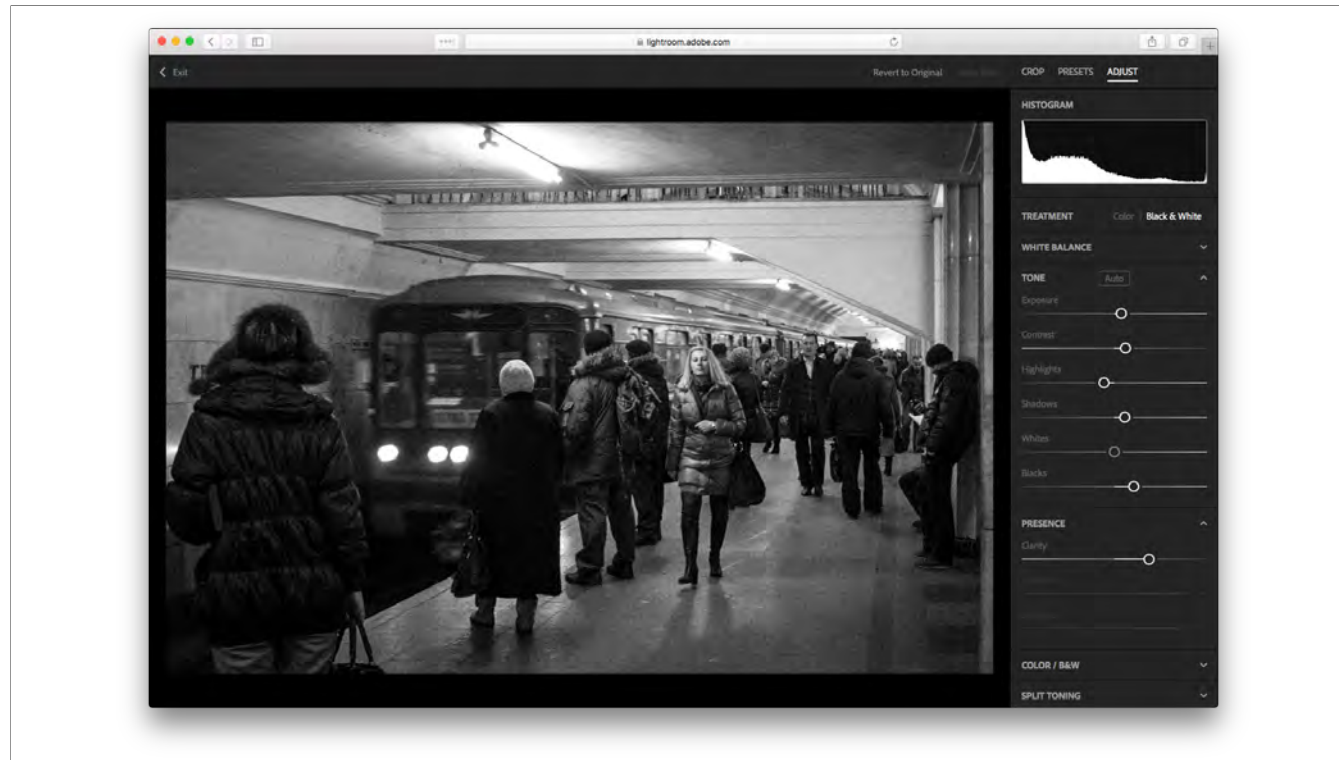


C++ Seasoning From Going Native 2013

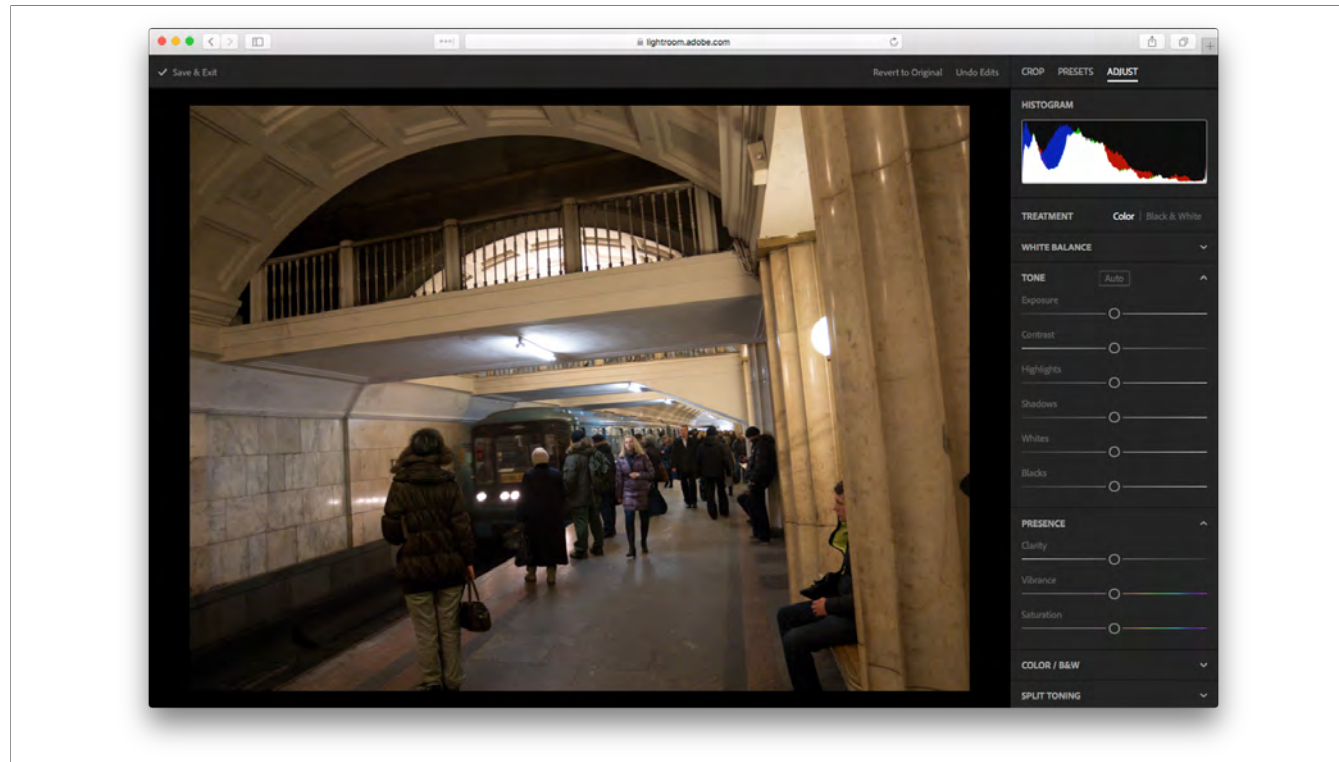
Common Themes

- Manage Relationships
- Understand the Fundamentals
- Code Simply
- Local and Equational Reasoning

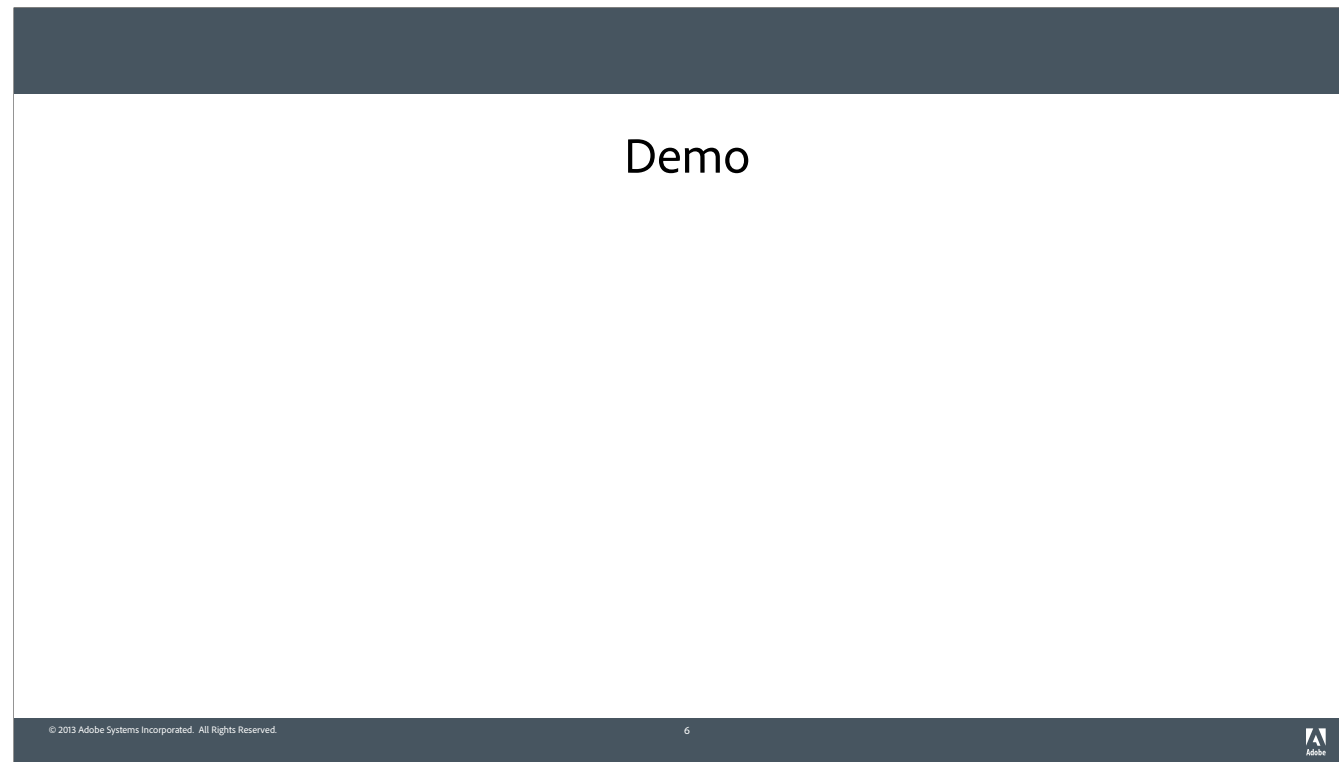
Code Simply - killing me for this talk...



Lr Web



Lr Web



LrM demo

Discussion of taking large, multi-threaded code base to run with Google PNaCl and compiled single threaded to JS. Give demo in Safari.

Concurrency

- Concurrency: when tasks start, run, and complete in overlapping time periods
- Parallelism: when two or more tasks execute simultaneously
- Why?
 - Enable performance through parallelism
 - Improve interactivity by handling user actions concurrent with processing and IO

<http://docs.oracle.com/cd/E19455-01/806-5257/ijep9032b/index.html>

Interactivity often impedes overall performance

Goal: No Raw Synchronization Primitives

What are raw synchronization primitives?

- Synchronization primitives are basic constructs such as:
 - Mutex
 - Atomic
 - Semaphore
 - Memory Fence
 - Condition Variable

You Will Likely Get It Wrong

Emailing Hans Boehm about about how to correctly communicate a bool for cancelation.

Problems with Locks

```
template <typename T>
class bad_cow {
    struct object_t {
        explicit object_t(const T& x) : data_m(x) {}
        atomic<int> count_m{1};
        T data_m; };
    object_t* object_m;
public:
    explicit bad_cow(const T& x) : object_m(new object_t(x)) { }
    ~bad_cow() { if (0 == --object_m->count_m) delete object_m; }
    bad_cow(const bad_cow& x) : object_m(x.object_m) { ++object_m->count_m; }

    bad_cow& operator=(const T& x) {
        if (object_m->count_m == 1) object_m->data_m = x;
        else {
            object_t* tmp = new object_t(x);
            --object_m->count_m;
            object_m = tmp;
        }
        return *this;
    }
};
```

- There is a subtle race condition here:
- if count != 1 then the bad_cow could also be owned by another thread(s)
- if the other thread(s) releases the bad_cow between these two atomic operations
- then our count will fall to zero and we will leak the object

I marked in red each of the atomic operations
There is a subtle bug in this code...

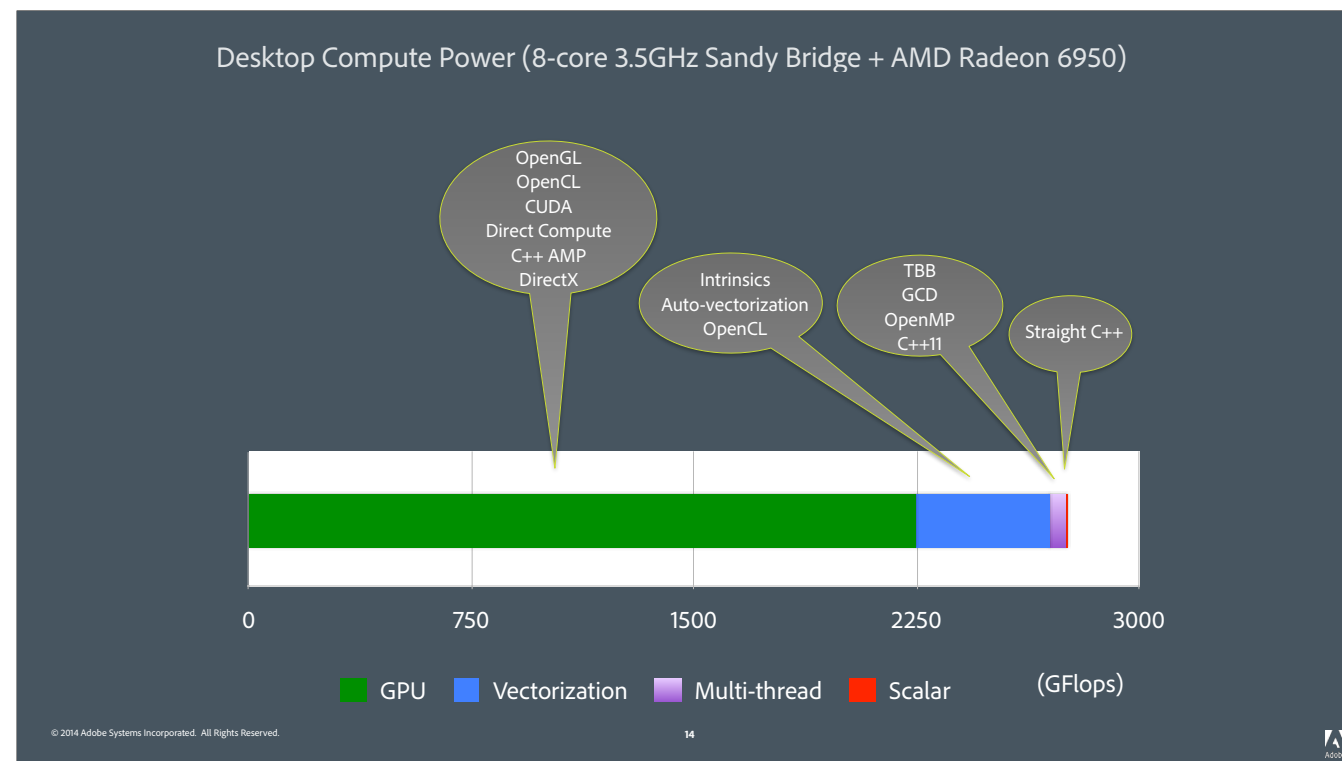
Problems with Locks

```
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        explicit object_t(const T& x) : data_m(x) {}
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        T data_m; };
    object_t* object_m;
public:
    explicit bad_cow(const T& x) : object_m(new object_t(x)) { }
    ~bad_cow() { if (0 == --object_m->count_m) delete object_m; }
    bad_cow(const bad_cow& x) : object_m(x.object_m) { ++object_m->count_m; }

    bad_cow& operator=(const T& x) {
        if (object_m->count_m == 1) object_m->data_m = x;
        else {
            object t* tmp = new object t(x);
            if (0 == --object_m->count_m) delete object_m;
            object_m = tmp;
        }
        return *this;
    }
};
```

I marked in red each of the atomic operations
There is a subtle bug in this code...

Performance through Parallelism



What does the available computing power in a modern high end machine look like?

“Guaranteed not to exceed” Card \$250. GPU: 2250, SSE: 500, thread: 50, scalar: 7, 0.25% of total, 1.5% of cpu. On-board CPU graphics are not included.

C++11 gets us to maybe 1% of the GFlops

Thread Building Blocks,

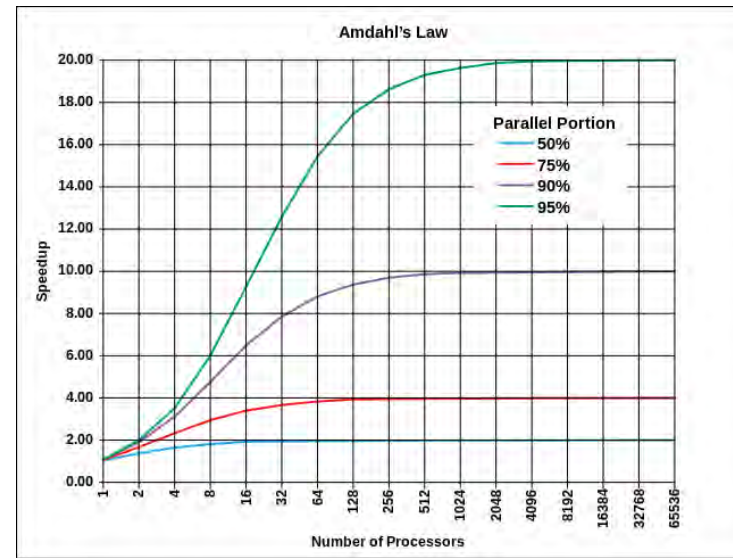
Apple Grand Central Dispatch

GPU is a latency-hiding, data-parallel, bandwidth-oriented world. CPU is a latency-sensitive, branching world.

Bandwidth 25 GB CPU / 180 GB GPU, 400-500 GB/s in 2014.

Amdahl's Law

$$S(N) = \frac{1}{(1 - P) + \frac{P}{N}}$$

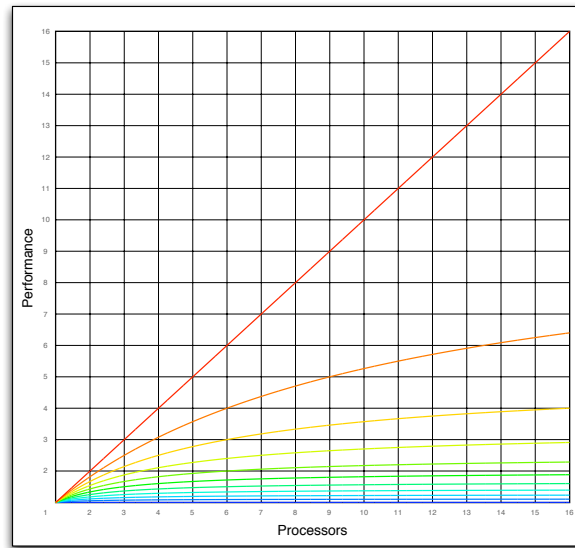


http://en.wikipedia.org/wiki/Amdahl%27s_Law

This is a horrible graph - logarithmic scale
Linear speed up arcs up to the moon (if printed on paper about 1/2 mile(or km) away).

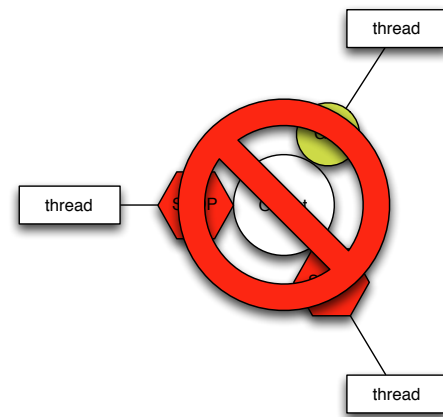
Amdahl's Law

Each line represents 10% more
synchronization



Code Simply - killing me for this talk...

Why No Raw Synchronization Primitives?





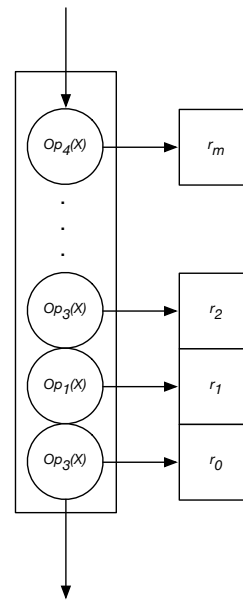
Mutexes and Sequential Consistency

```
class registry {  
    mutex _mutex;  
    unordered_map<string, string> _map;  
public:  
    void set(string key, string value) {  
        unique_lock<mutex> lock(mutex);  
        _map.emplace(move(key), move(value));  
    }  
  
    auto get(const string& key) -> string {  
        unique_lock<mutex> lock(mutex);  
        return _map.at(key);  
    }  
};
```

"It can be shown that programs that correctly use mutexes and `memory_order_seq_cst` operations to prevent all data races and use no other synchronization operations behave as if the operations executed by their constituent threads were simply interleaved, with each value computation of an object being taken from the last side effect on that object in that interleaving. This is normally referred to as '*sequential consistency*'."

– C++11 Standard 1.10.21

Mutexes and Sequential Consistency



Interactivity often impedes overall performance

Mutexes and Sequential Consistency

- A mutex serializes a set of operations, Op_n , where the operation is the code executed while the mutex is locked
- Operations are interleaved and may be executed in any order and may be repeated
- Each operation takes an argument, X , which is the set of all objects mutated under all operations
 - X may not be safely read or written without holding the lock if it may be modified by a task holding the lock
- Each operation may yield a result, r_m , which can communicate information about the state of X while it's associated operation was executed
- The same is true of all atomic operations

Mutexes and Sequential Consistency

```
class registry {  
    serial_queue _q;  
  
    using map_t = unordered_map<string, string>;  
  
    shared_ptr<map_t> _map = make_shared<map_t>();  
public:  
    void set(string key, string value) {  
        _q.async([_map = _map](string key, string value) {  
            _map->emplace(move(key), move(value));  
        }, move(key), move(value));  
    }  
  
    auto get(string key) -> future<string> {  
        return _q.async([_map = _map](string key) {  
            return _map->at(key);  
        }, move(key));  
    }  
};
```


Mutexes and Sequential Consistency

```
class registry {
    serial_queue _q;

    using map_t = unordered_map<string, string>;

    shared_ptr<map_t> _map = make_shared<map_t>();
public:
    void set(string key, string value) {
        _q.async([_map = _map](string key, string value) {
            _map->emplace(move(key), move(value));
        }, move(key), move(value));
    }

    auto get(string key) -> future<string> {
        return _q.async([_map = _map](string key) {
            return _map->at(key);
        }, move(key));
    }

    void set(vector<pair<string, string>> sequence) {
        _q.async([_map = _map](vector<pair<string, string>> sequence) {
            _map->insert(make_move_iterator(begin(sequence)), make_move_iterator(end(sequence)));
        }, move(sequence));
    }
};
```


Mutexes and Sequential Consistency

- The transformation mutex to serial queue places an upper-bound
 - Synchronization overhead
 - Time to issue operation

Threads and Tasks

- Thread: Execution environment consisting of a stack and processor state running in parallel to other threads
- Task: A unit of work, often a function, to be executed on a thread
- Tasks are scheduled on a thread pool to optimize machine utilization

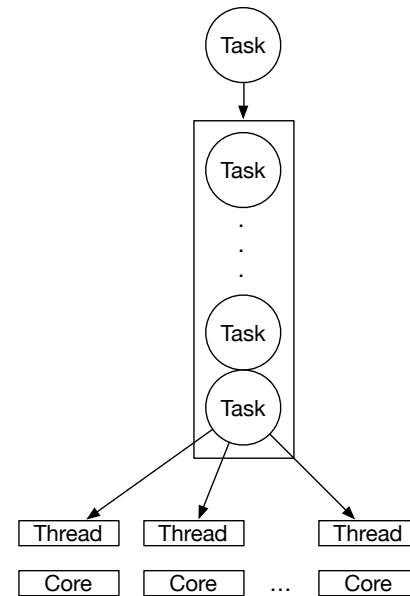
Interactivity often impedes overall performance

- C++14 does not (really) have a task system
 - Threads
 - Futures
- It is implementation defined if `std::async()` spins up a thread or executes on a thread pool.

Building a Task System

- Portable Reference Implementation in C++14
- Windows - Window Thread Pool and PPL
- Apple - Grand Central Dispatch (libdispatch)
 - open source, runs on Linux and Android
- Intel TBB - many platforms
 - open source
- HPX - many platforms
 - open source

Building a Task System



<http://docs.oracle.com/od/E19233-01/B16-5137/ggadu/index.html>

We have a single queue
Similar to Oracle multithreaded programming guide

Building a Task System

```
using lock_t = unique_lock<mutex>;

class notification_queue {
    deque<function<void()>> _q;
    mutex _mutex;
    condition_variable _ready;

public:
    void pop(function<void()>& x) {
        lock_t lock{_mutex};
        while (_q.empty()) _ready.wait(lock);
        x = move(_q.front());
        _q.pop_front();
    }

    template<typename F>
    void push(F&& f) {
        {
            lock_t lock{_mutex};
            _q.emplace_back(forward<F>(f));
        }
        _ready.notify_one();
    }
};
```

Building a Task System

```
class task_system {
    const unsigned _count{thread::hardware_concurrency()};
    vector<thread> _threads;
    notification_queue _q;

    void run(unsigned i) {
        while (true) {
            function<void()> f;
            _q.pop(f);
            f();
        }
    }

public:
    task_system() {
        for (unsigned n = 0; n != _count; ++n) {
            _threads.emplace_back([&, n]{ run(n); });
        }
    }

    ~task_system() {
        for (auto& e : _threads) e.join();
    }

    template <typename F>
    void async_(F&& f) {
        _q.push(forward<F>(f));
    }
};
```


Building a Task System

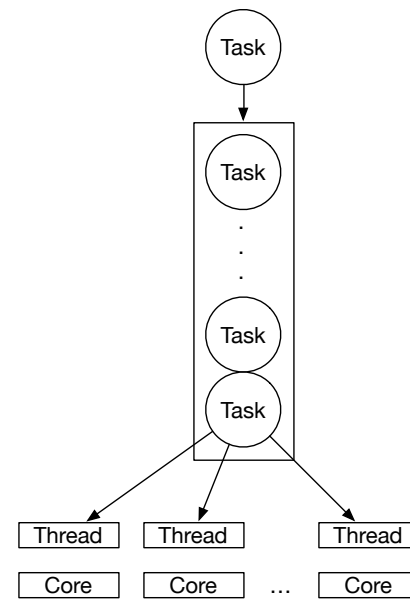
```
class notification_queue {
    deque<function<void()>> _q;
    bool _done{false};
    mutex _mutex;
    condition_variable _ready;

public:
    void done() {
        {
            unique_lock<mutex> lock{_mutex};
            _done = true;
        }
        _ready.notify_all();
    }

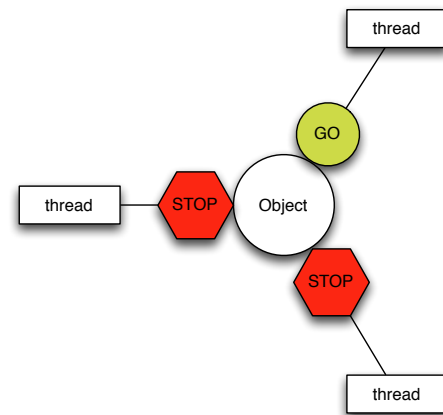
    bool pop(function<void()>& x) {
        lock_t lock{_mutex};
        while (_q.empty() && !_done) _ready.wait(lock);
        if (_q.empty()) return false;
        x = move(_q.front());
        _q.pop_front();
        return true;
    }

    template<typename F>
    void push(F&& f) {
        {
            lock_t lock{_mutex};
            q.emplace_back(forward<F>(f));
        }
    }
};
```

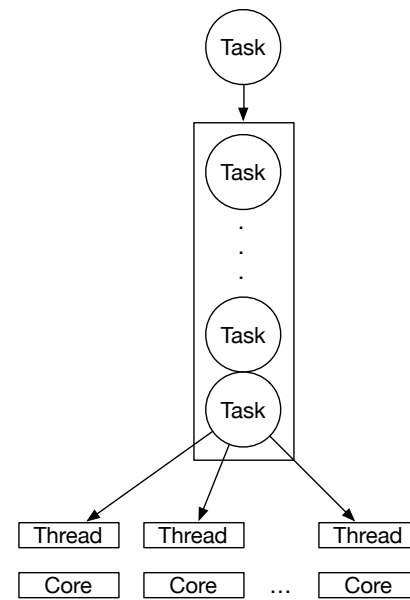

Building a Task System



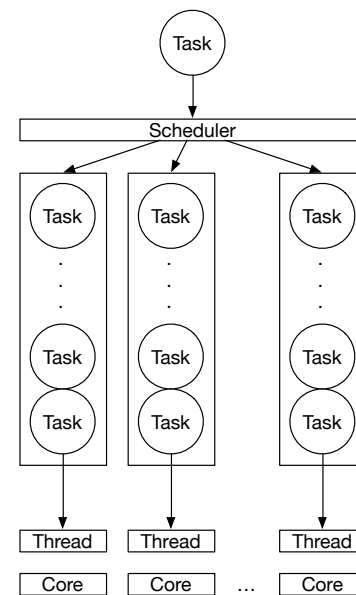
Why No Raw Synchronization Primitives?



Building a Task System



Building a Task System



Building a Task System

```
class task_system {
    const unsigned _count{thread::hardware_concurrency()};
    vector<thread> _threads;
    vector<notification_queue> _q{_count};
    atomic<unsigned> _index{0};

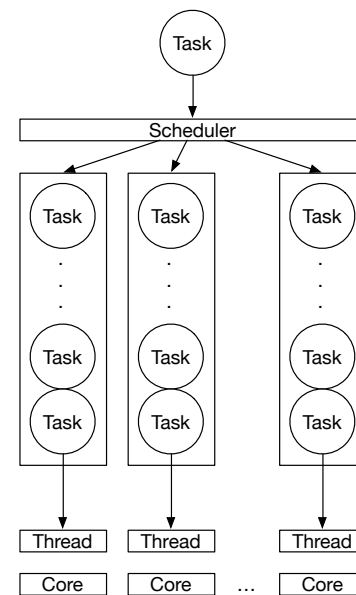
    void run(unsigned i) {
        while (true) {
            function<void()> f;
            if (!_q[i].pop(f)) break;
            f();
        }
    }

public:
    task_system() {}

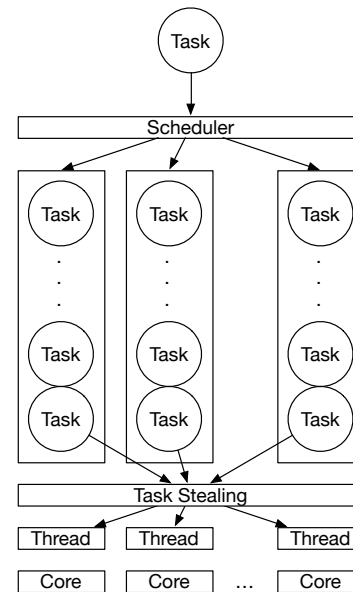
    ~task_system() {
        for (auto& e : _q) e.done();
        for (auto& e : _threads) e.join();
    }

    template <typename F>
    void async_(F&& f) {
        auto i = _index++;
        _q[i % _count].push(forward<F>(f));
    }
};
```

Building a Task System



Building a Task System



Building a Task System

```
class notification_queue {
    deque<function<void()>> _q;
    bool _done{false};
    mutex _mutex;
    condition_variable _ready;

public:
    bool try_pop(function<void()>& x) {
        lock_t lock{_mutex, try_to_lock};
        if (!lock || _q.empty()) return false;
        x = move(_q.front());
        _q.pop_front();
        return true;
    }

    template<typename F>
    bool try_push(F&& f) {
        {
            lock_t lock{_mutex, try_to_lock};
            if (!lock) return false;
            _q.emplace_back(forward<F>(f));
        }
        _ready.notify_one();
        return true;
    }

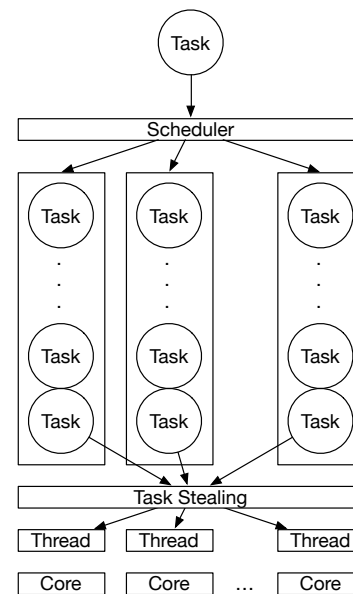
    void done() {
        {
            unique_lock<mutex> lock{ _mutex};
```


Building a Task System

```
void run(unsigned i) {  
    while (true) {  
        function<void()> f;  
  
        for (unsigned n = 0; n != _count; ++n) {  
            if (_q[(i + n) % _count].try_pop(f)) break;  
        }  
        if (!f && !_q[i].pop(f)) break;  
  
        f();  
    }  
}  
  
public:  
    task_system() { }  
    ~task_system() { }  
  
    template <typename F>  
    void async_(F&& f) {  
        auto i = _index++;  
  
        for (unsigned n = 0; n != _count * K; ++n) {  
            if (_q[(i + n) % _count].try_push(forward<F>(f))) return;  
        }  
  
        _q[i % _count].push(forward<F>(f));  
    }  
};
```

K is about 56.

Building a Task System



- Compared to Apple's Grand Central Dispatch (libdispatch)



C++14 compatible async with libdispatch

```
template <class Function, class... Args>
auto async(Function&& f, Args&&... args )
{
    using result_type = std::result_of_t<std::decay_t<Function>(std::decay_t<Args>...)>;
    using packaged_type = std::packaged_task<result_type>;

    auto _p = new packaged_type(std::bind([_f = std::forward<Function>(f)](Args&... args) {
        return _f(std::move(args)...);
    }, std::forward<Args>(args)...));

    auto result = _p->get_future();

    dispatch_async_f(dispatch_get_global_queue(DISPATCH_QUEUE_PRIORITY_DEFAULT, 0),
        _p, [](void* p) {
            auto _p = static_cast<packaged_type*>(p);
            (*_p)();
            delete _p;
        });

    return result;
}
```

Note use of bind and lambda to forward arguments so move only arguments can be correctly passed through.

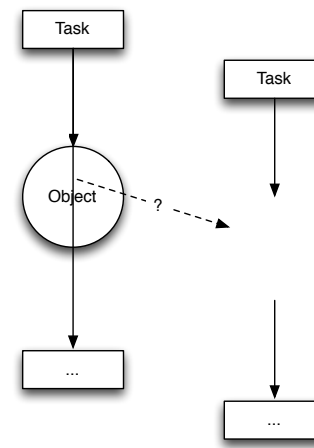
Task System

- Written with ASIO (Boost 1.62.0)

```
class task_system {  
    io_service          _service;  
    vector<thread>      _threads;  
    unique_ptr<io_service::work> _work{make_unique<io_service::work>(_service)};  
  
public:  
    task_system() {  
        for (unsigned n = 0; n != thread::hardware_concurrency(); ++n) {  
            _threads.emplace_back([&]{  
                _service.run();  
            });  
        }  
    }  
  
    ~task_system() {  
        _work.reset();  
        for (auto& e : _threads) e.join();  
    }  
  
    template <typename F>  
    void async_(F&& f) {  
        _service.post(forward<F>(f));  
    }  
};
```



No Raw Synchronization Primitives



This is a communicating sequential process model

Futures

```
future<cpp_int> x = async([]{ return fibonacci<cpp_int>(1'000'000); });  
// Do Something  
cout << x.get() << endl;
```

- Fibonacci is often used as an example for parallel algorithms
 - Please stop...

cpp_int is boost arbitrary precision integer

Public Service Announcement - How to Write Fibonacci

```
template <typename T, typename N, typename Op>
T power(T x, N n, Op op)
{
    if (n == 0) return identity_element(op);

    while ((n & 1) == 0) {
        n >>= 1;
        x = op(x, x);
    }

    T result = x;
    n >>= 1;
    while (n != 0) {
        x = op(x, x);
        if ((n & 1) != 0) result = op(result, x);
        n >>= 1;
    }
    return result;
}
```

Egyptian Multiplication (Russian Peasant Algorithm)

See "From Mathematics to Generic Programming" - Alex Stepanov and Dan Rose

Public Service Announcement - How to Write Fibonacci

```
template <typename N>
struct multiply_2x2 {
    array<N, 4> operator()(const array<N, 4>& x, const array<N, 4>& y)
    {
        return { x[0] * y[0] + x[1] * y[2], x[0] * y[1] + x[1] * y[3],
                 x[2] * y[0] + x[3] * y[2], x[2] * y[1] + x[3] * y[3] };
    }
};

template <typename N>
array<N, 4> identity_element(const multiply_2x2<N>&) { return { N(1), N(0), N(0), N(1) }; }

template <typename R, typename N>
R fibonacci(N n) {
    if (n == 0) return R(0);
    return power(array<R, 4>{ 1, 1, 1, 0 }, N(n - 1), multiply_2x2<R>())[0];
}
```

$$\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}^n = \begin{bmatrix} F_{n+1} & F_n \\ F_n & F_{n-1} \end{bmatrix}$$

Fibonacci is a linear occurrence

Futures

```
future<cpp_int> x = async([]{ return fibonacci<cpp_int>(1'000'000); });  
// Do Something  
cout << x.get() << endl;
```

cpp_int is boost arbitrary precision integer

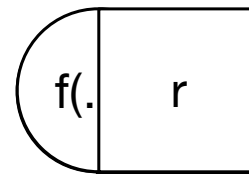
Futures

19532821287077577316320149475962563324435429965918733969534051945716252578870156947666419876341501461288795
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99360209593392020051848620284024473431398113674187202038684801753185386211128781082406177413832935545616876
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28124660628996787166529678487268484905041328497297712688011639978376434280202452251550102240354169885185375
01584673881194047619720619603126534496759917893244478170702904446589571950228809157793897642423751814020998
00958161231477902295781100168670186738619861797138139854666281969548553740707356228616165539428076418408092
12047932816683005984504787929406356318097479755152035094682765918741610907637506902765294367561539803261388
001044850410045202754188004573562070542180662063441346306055080091375010053760250440113617210176881147264

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3.5s to calculate
208,988 digits



Futures

- Futures allow minimal code transformations to express dependencies

[future vs. callback]

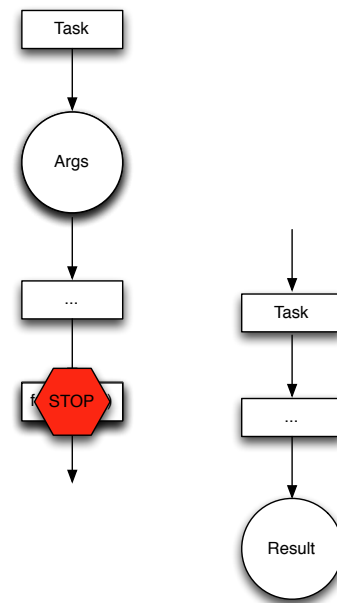
Exception Marshalling

```
future<cpp_int> x = async([]{  
    throw runtime_error("failure");  
    return fibonacci<cpp_int>(1'000'000);  
});  
  
// Do Something  
  
try {  
    cout << x.get() << endl;  
} catch (const runtime_error& error) {  
    cout << error.what() << endl;  
}
```

failure

Error type and value preserved

No Raw Synchronization Primitives

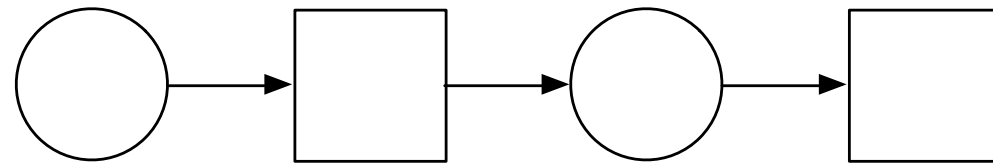


This is a communicating sequential process model

Futures: What year is this?

- C++11 (through 17) futures lack:
 - Continuations - `.then()`
 - Joins - `when_all()`
 - Split
 - Cancellation
 - Progress Monitoring (Except Ready)
- And C++11 futures don't compose (easily) to add these features

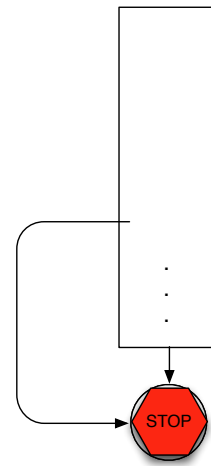
d



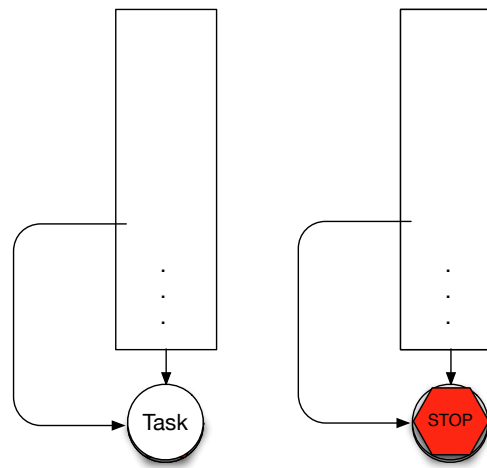
Futures: Continuations

- Blocking on `std::future.get()` has two problems
 - One thread resource is consumed, increasing contention
 - Possibly causing a deadlock in our tasking system!
 - Any subsequent non-dependent calculations on the task are also blocked
- C++17 doesn't have continuations
 - GCD has serialized queues and groups
 - PPL has chained tasks
 - TBB has flow graphs
 - TS Concurrency has `.then()`
 - Boost futures have them now

Futures: get() deadlock



Futures: get() deadlock



- Blocking on `std::future.get()`
 - Very difficult to use safely with a thread pool
 - C++11 allows `std::async()` to use a thread pool
 - But the futures must contain a task stealing `get()` operation to make this work
 - You cannot create an `std::future()` with this attribute except with `std::async()`
- Not just `get()` - *any* conditional blocking (condition variables, wait, ...) is problematic with a task system

Do not call `std::future.get()` or `std::future.wait()` when the originating task, or any subordinate task, is on the same queue, even if it is a concurrent queue

Important: You should never call the `dispatch_sync` or `dispatch_sync_f` function from a task that is executing in the same queue that you are planning to pass to the function. This is particularly important for serial queues, which are guaranteed to deadlock, but should also be avoided for concurrent queues.

<https://developer.apple.com/library/content/documentation/General/Conceptual/ConcurrencyProgrammingGuide/OperationQueues/OperationQueues.html>

Futures: Continuations

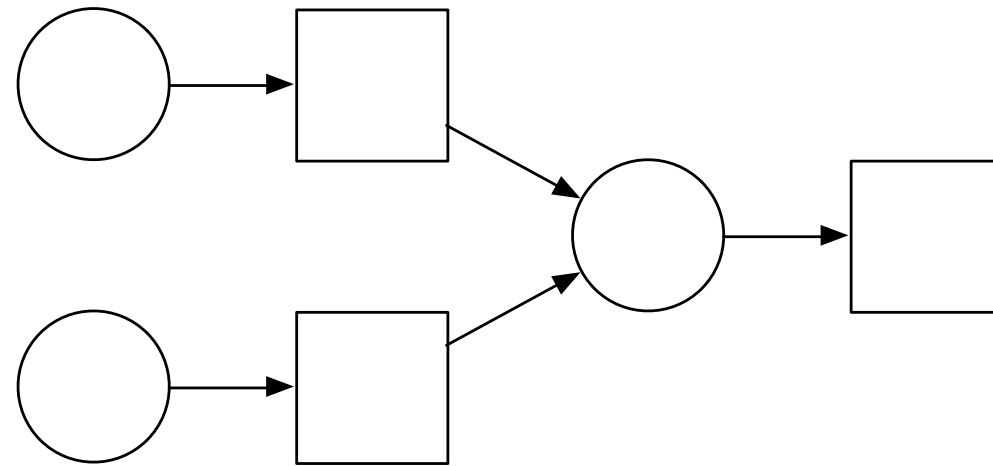
```
future<cpp_int> x = async([]{ return fibonacci<cpp_int>(1'000); });  
future<void> y = x.then([](future<cpp_int> x){ cout << x.get() << endl; });  
  
// Do something  
  
y.wait();
```

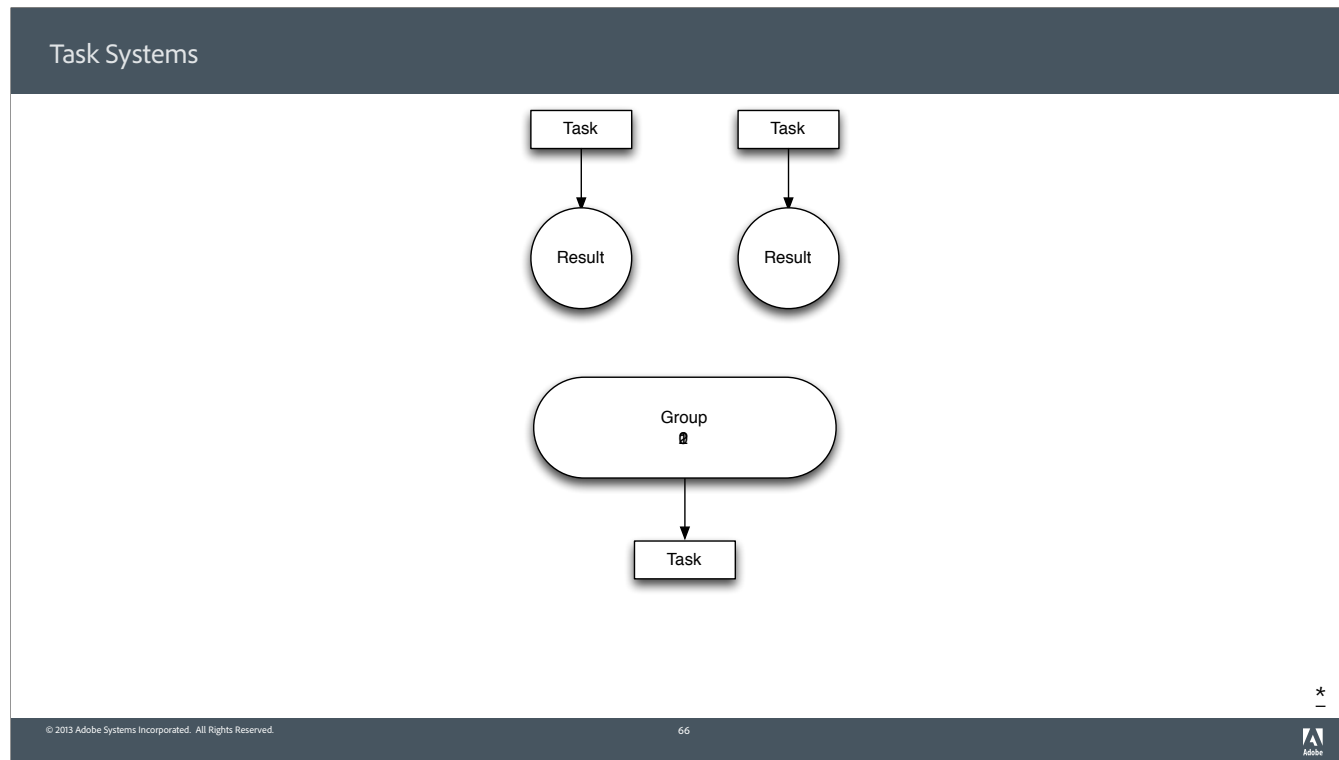
43466557686937456435688527675040625802564660517371780402481729089536555417949051890403879840079255169295922593080322634775209
689623239873322471161642996440906533187938298969649928516003704476137795166849228875

Futures vs Completion Handlers

- Completion handlers are callbacks, they must be known prior to the call
 - No need to synchronize between invoking and setting the continuation
- Futures allow setting the continuation after the sending call is in flight
 - Simpler to compose
 - Require synchronization between invoking and setting the continuation

A low-level tasking system as we built can be adapted to either - see libdispatch slide





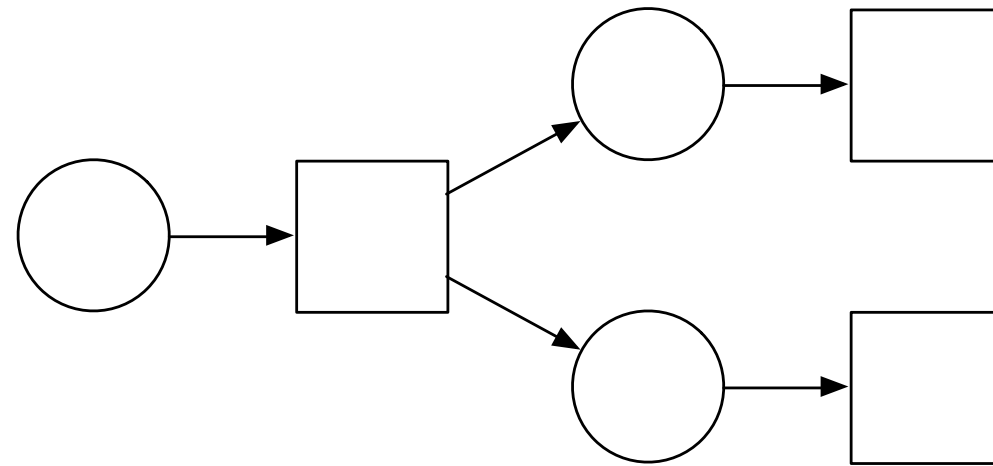
Add seasoning slide with packaged task
Little discussion on other items are too complex.

Futures: Continuations

```
auto x = async([]{ return fibonacci<cpp_int>(1'000'000); });  
auto y = async([]{ return fibonacci<cpp_int>(2'000'000); });  
  
auto z = when_all(std::move(x), std::move(y)).then([](auto f){  
    auto t = f.get();  
    return cpp_int(get<0>(t).get() * get<1>(t).get());  
});  
  
cout << z.get() << endl;
```

f is a future tuple of futures

result is 626,964 digits



Futures: Continuations

```
future<cpp_int> x = async([]{ return fibonacci<cpp_int>(100); });  
future<cpp_int> y = x.then([](future<cpp_int> x){ return cpp_int(x.get() * 2); });  
future<cpp_int> z = x.then([](future<cpp_int> x){ return cpp_int(x.get() / 15); });
```

Thread 1: signal SIGABRT

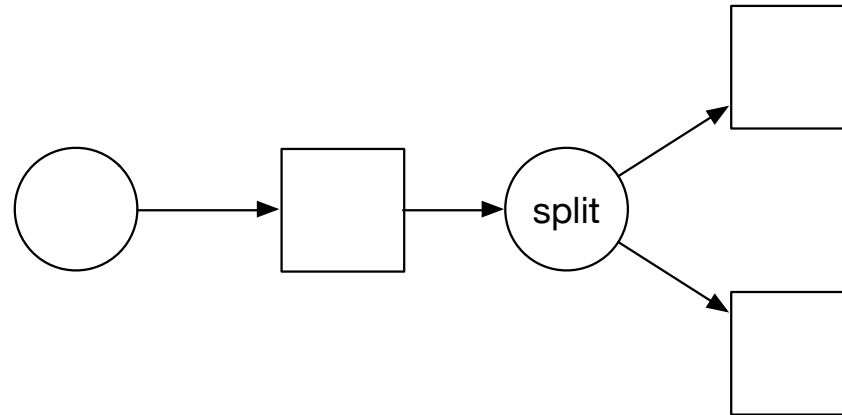
Assertion failed: (px != 0), function operator->, file shared_ptr.hpp, line 648.

If I x is share() – it doesn't crash but second then() overwrites the first.

Continuations

- Desired behavior
 - A future should behave as a *regular* type - a token for the actual value
 - `shared_futures` let me “copy” them around and do multiple `get ( )` operations
 - But not multiple continuations (at least with Boost as of 1.62, the TS is ambiguous here!)

- We can write a pseudo-copy, `split()`.



Kleisli Category

Futures: Continuations

```
future<cpp_int> x = async([]{ return fibonacci<cpp_int>(100); });  
  
future<cpp_int> y = split(x).then([](future<cpp_int> x){ return cpp_int(x.get() * 2); });  
future<cpp_int> z = x.then([](future<cpp_int> x){ return cpp_int(x.get() / 15); });  
  
future<void> done = when_all(std::move(y), std::move(z)).then([](auto f){  
    auto t = f.get();  
    cout << get<0>(t).get() << endl;  
    cout << get<1>(t).get() << endl;  
});  
  
done.wait();
```

708449696358523830150
23614989878617461005

Split x

Building Blocks

- Promise is the sending side of a future
- Promises are packaged with a function to form a *packaged task*
 - Packaged tasks handle the exception marshalling through a promise

Promise

```
promise<int> x;  
future<int> y = x.get_future();  
  
x.set_value(42);  
cout << y.get() << endl;
```

42

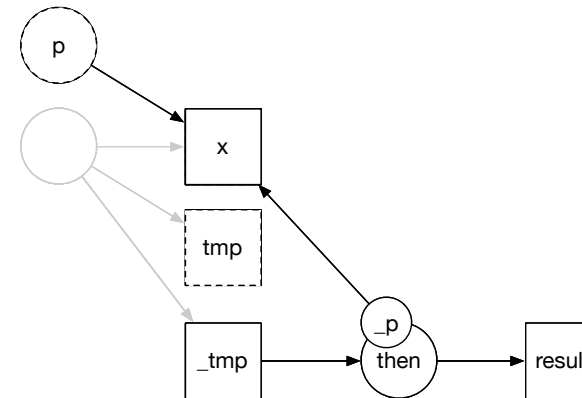
Use specialization to extract parts of function signature
Can use boost function traits

Futures: Split

```
template <typename T>
auto split(future<T>& x) {
    auto tmp = std::move(x);

    promise<T> p;
    x = p.get_future(); // replace x with new future

    return tmp.then([&p = move(p)](auto _tmp) mutable {
        auto value = _tmp.get();
        _p.set_value(value); // assign to new "x" future
        return value; // return value through future result
    });
}
```



Use specialization to extract parts of function signature
Can use boost function traits

[*] Animate (build in) – diagram? circle for promise with arrow to future.

Futures: Split

```
template <typename T>
auto split(future<T>& x) {

    auto tmp = std::move(x);

    promise<T> p;
    x = p.get_future(); // replace x with new future

    return tmp.then([&p = std::move(p)](auto _tmp) mutable {
        if (_tmp.has_exception()) {
            auto error = _tmp.get_exception_ptr();
            _p.set_exception(error);
            rethrow_exception(error);
        }

        auto value = _tmp.get();
        _p.set_value(value); // assign to new "x" future
        return value; // return value through future result
    });
}
```

Use specialization to extract parts of function signature
Can use boost function traits

Futures: Continuations

```
future<cpp_int> x = async([]{ return fibonacci<cpp_int>(100); });  
  
future<cpp_int> y = split(x).then([](future<cpp_int> x){ return cpp_int(x.get() * 2); });  
future<cpp_int> z = x.then([](future<cpp_int> x){ return cpp_int(x.get() / 15); });  
  
future<void> done = when_all(std::move(y), std::move(z)).then([](auto f){  
    auto t = f.get();  
    cout << get<0>(t).get() << endl;  
    cout << get<1>(t).get() << endl;  
});  
  
done.wait();
```

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23614989878617461005

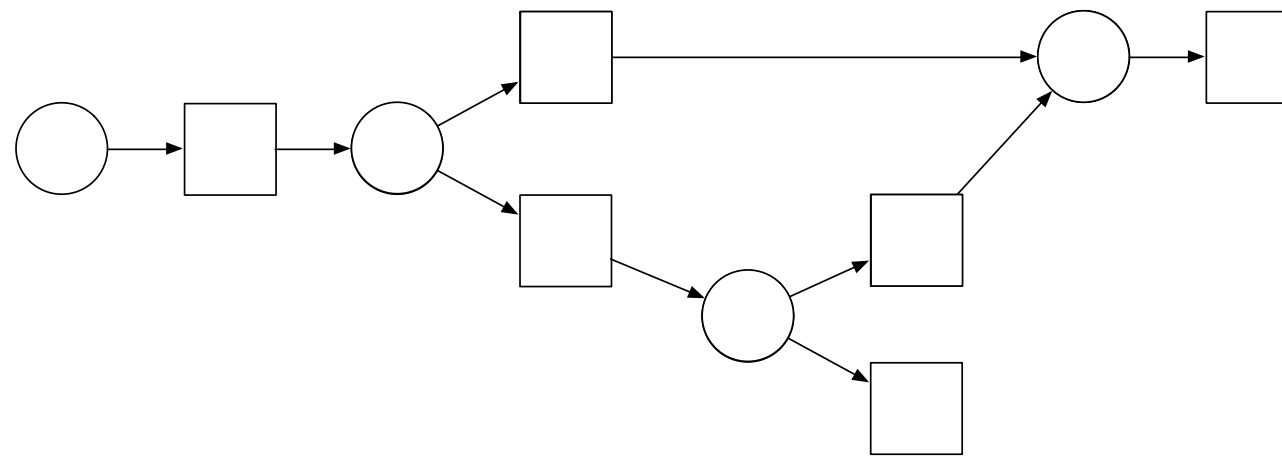
Split x

Cancellation

- When the (last) future destructs
 - The associated task that has not started, should not execute (NOP)
 - The resource held by that task should be released
 - Since that task may hold futures for other tasks, the system unravels
- I do not know of a good way to compose such cancellation with current futures

For shared futures, otherwise when “the future”
[*] Reference stlab/libraries? PPL cancellation?

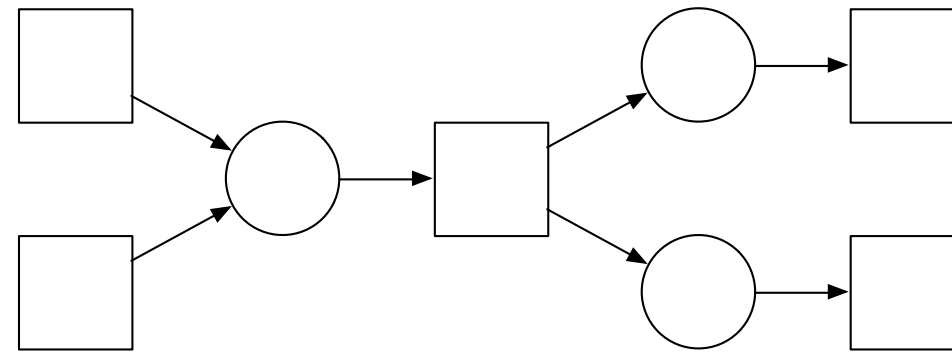
Cancellation



- Currently supports
 - Multiple continuations and copy
 - Optimized for rvalues
 - Join (When All, When Any)
 - Cancellation on Destruction (and explicit reset)
 - And detach
- <https://github.com/stlab/libraries/tree/develop>
- Thanks to Felix Petriconi

Channels

What if we persist the graph?



What if we persist the graph?

- Allow multiple invocations of the tasks by setting the source values
- Each change triggers a notification to the sink values
 - This is a reactive programming model and futures are known as *behaviors* or *channels*

Accumulators and Generator

- Each operation does not have to be a 1:1 mapping of input to output
- Coroutines are one way to write n:m functions

Channels

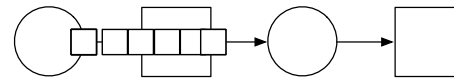
```
channel<int> send;

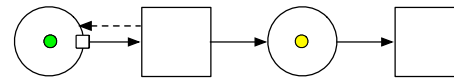
auto hold = send
| [](const receiver<int>& r) {
    int sum = 0;
    while(auto v = co_await r) {
        sum += v.get();
    }
    return sum;
}
| [](int x){ cout << x << '\n'; };

send(1);
send(2);
send(3);
send.close();
```

6

[insert slide here on flow control]





Channels

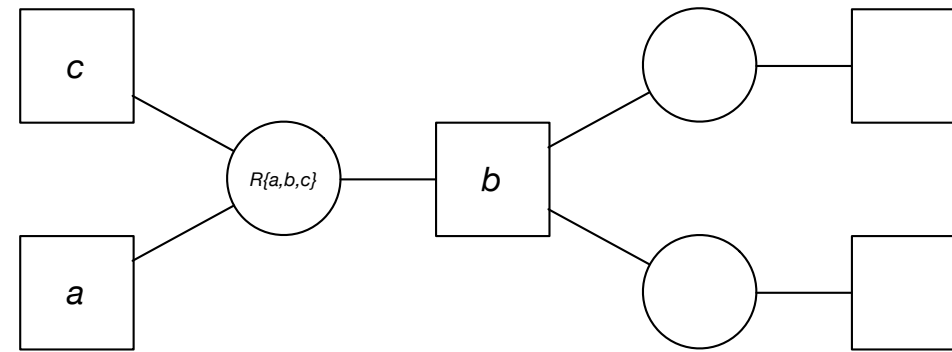
```
task<void> render(channel<parameters> in) {  
    while (true) {  
        optional<parameters> params;  
        while (params = co_await in) {  
            co_yield render_frame(*params, draft_quality);  
        }  
        co_yield render_frame(*params, final_quality);  
    }  
}
```

[insert slide here on flow control]

Property Models

How do the graphs change during execution?

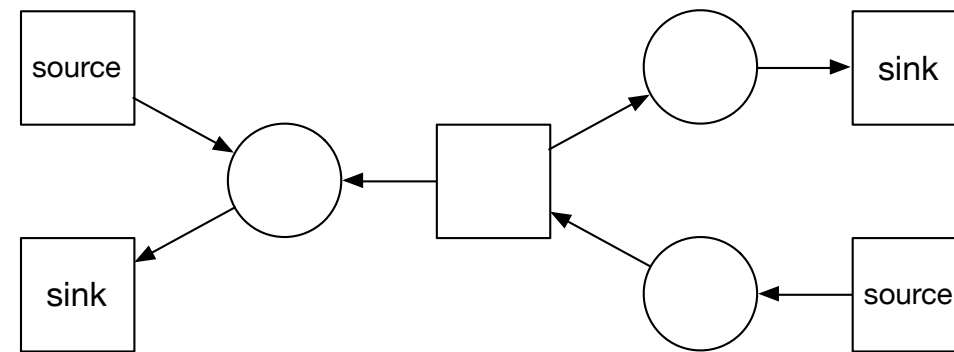
Property Model



A function is a directed relationship

- We can remove the arrows by providing a package of functions to represent the relationship
 - $a = b * c$
 $b = a / c$
 $c = a / b$
 - This forms a type of constraint system called a *property model*
 - Flow is determined by value, or *cell*, priority
 - Relationships can be conditional, so long as predicate can be determined regardless of flow
- Cells can only have one in-edge for a given flow or the system is over constrained

Property Model

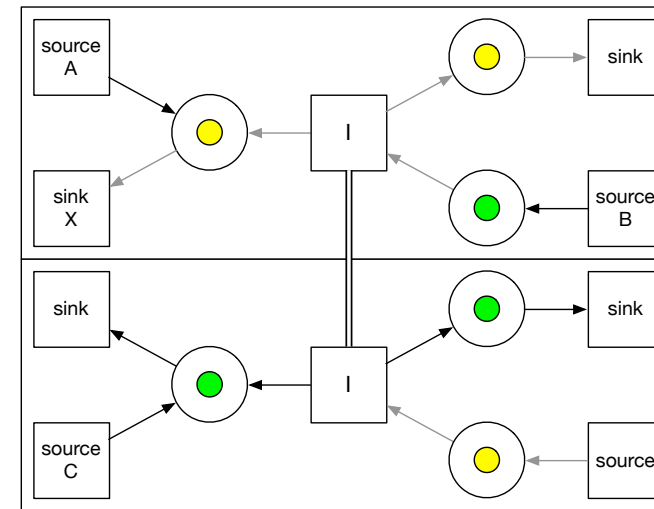


Property Models

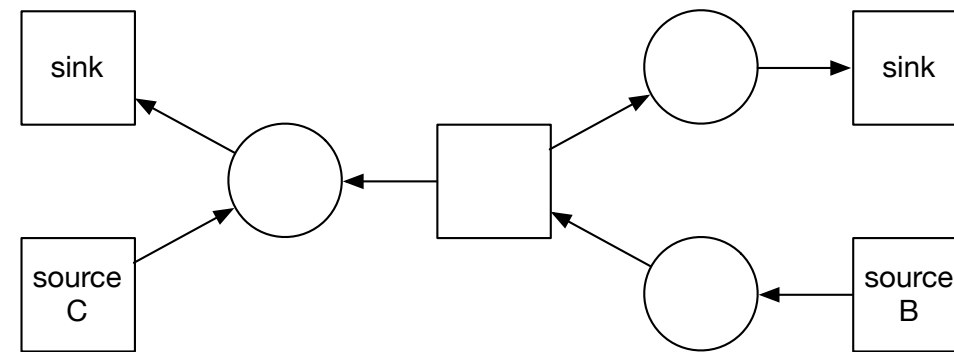
- Reflowing a property model doesn't require all relationships to be resolved
 - The task representing them can still be executing concurrently
- This creates a single dependency graph that is appended to for each new flow and is pruned and *unravels* as tasks are complete

Property Model

- Value set in source A then B
- Operation connected to B is high latency
- Value is set in source C causing reflow
 - Sink X is no longer needed, pending operation is canceled
 - Source A is discarded
 - Intermediate value I is shared between flows, once determined
- Final values determined by source B (via I) and source C



Property Model



- Very useful for UI behavior
- Significant information is in the graph
 - Source / Derived values form a partition set
 - Easily model checked
 - Equal result regardless of source order
 - Form an operational transform, useful for collaborative editing
 - A value is *implied* by the current state it only has in in-edge
 - Source values determine *intent*
 - Values disconnected from result (sink) are *don't care*

Final Thoughts

- Perhaps representing such systems *as if* it where imperative code is not the correct approach
- Instead a graph description can be compiled and statically validated
- Slides and code from talk (and property model papers):
 - <http://sean-parent.stlab.cc/papers-and-presentations>
- Experimental future and channel library:
 - <https://github.com/stlab/libraries/tree/develop>
 - Thanks to Felix Petriconi
- Communicating Sequential Processes (C. A. R. Hoare)
 - <http://usingcsp.com/cspbook.pdf>

